

Predicting Contract Renewal in Public Digital Procurement

A Survival-Analysis Approach on BOAMP Notices for Gigalis
Internship Report — Executive Orientation

Internship Report — Gigalis (Pays de la Loire)

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Abstract

This is the **executive report** of the internship: it lets a reader scan the project rapidly and understand *what was done*, *why it matters*, and *what is reliable*. The full implementation and diagnostics are in the **technical report** (`phase1_technical_report.pdf`); the data reliability and event-definition detail are in the **data quality report**.

The work concerns predictive analysis of public digital procurement for **Gigalis**, a central purchasing body in Pays de la Loire. The goal is to move from a descriptive view (“what happened?”) towards a predictive one (“when will a contract be re-tendered?”). Because BOAMP contains *no explicit renewal field*, the renewal relationship is reconstructed from observable signals: a proxy renewal **event** built from Sentence-Transformer text embeddings combined with CPV, temporal, and buyer signals, used as the event variable for survival analysis. On 1,210 eligible source contracts the linking method recovers a plausible renewal for 665 of them (**55.0%**), against a 7.6% lexical baseline at the same window setting. Survival modeling gives a global Kaplan–Meier median survival of **50.1 months**, a Cox concordance of **0.6317**, and a best-fitting log-normal parametric model (AIC 7,068.4). The fitted model is translated into 12- and 24-month renewal-risk indicators at contract, buyer, and segment level. Buyer identities were consolidated by a SIREN enrichment step (completed 2026-06-24). The headline caveat: **event** is a proxy, and a manual audit of a stratified 150-case sample (2026-07-02) estimates its precision at only ≈ 9 –15% overall — concentrated in the high-text-similarity core (precision 1.00 at text ≥ 0.80 , ≈ 0.50 in the HIGH/strict tier, ≈ 0 below composite 0.50). The survival results should be read as describing algorithm-identifiable same-buyer re-publications, not validated renewals.

Calibration update — 2026-07-05

After the manual-audit finding that the broad baseline was too permissive, the event definition was recalibrated with a synthetic BOAMP benchmark and real BOAMP diagnostics. The synthetic benchmark is scored with the *same Sentence-Transformer encoder as the real pipeline* (`paraphrase-multilingual-MiniLM-L12-v2`). The recommended main rule is the **balanced** definition: text similarity ≥ 0.50 , composite score ≥ 0.50 , temporal window $W = 6$, corrected generic CPV scoring, and no margin floor. It produces 269 proxy recurrence events among 1,210 eligible source contracts (22.2%). The broad sensitivity gives 490 events (40.5%); the strict sensitivity gives 79 events (6.5%, underpowered for Cox/AFT).

The calibrated survival-ready file is `data/processed/boamp_phase2_survival_calibrated_balanced.csv`. Its Kaplan–Meier median is not reached; survival is 95.9% at 12 months, 92.3% at 24 months, 83.3% at 48 months, and 75.2% at 60 months. The Cox C-index is 0.592, and LogNormalAFT is the best balanced-rule AFT model by AIC (3,544.8). Re-scored 12/24-month risk indicators under this

rule give a mean 12-month recurrence probability of 0.023 and 27.5 expected recurrences within 12 months (`reports/tables/survival/renewal_risk_12_24_months_calibrated_balanced.csv`). These labels remain proxy recurrences, not legally verified renewals.

1 Context and Objective

1.1 Why this project

Gigalis operates a central purchasing body for digital services in Pays de la Loire. To anticipate re-tendering workloads, it wants to predict *when* digital contracts come up for renewal. The core obstacle: BOAMP publishes calls for tenders and awards, but *no notice says “this renews contract X”*. The renewal link must be inferred from text, classification, timing, and buyer identity.

1.2 Objective

The internship objective is to build a transparent, reproducible pipeline that (i) reconstructs a proxy renewal **event** from BOAMP signals, (ii) uses it as the outcome of a survival model of contract lifetime, and (iii) translates the model into operational renewal-risk indicators. This report orients the reader to that pipeline and its main results; exact field logic, formulas, and diagnostics are deferred to the technical and data quality reports.

2 Data Sources and Scope

2.1 Scope

- **Sectors:** CPV divisions 48 (software), 72 (IT services), 32 (telecom), 35 (security).
- **Territory:** departments 44, 49, 53, 72, 85 (Pays de la Loire).
- **Period:** 2015–2024. Study end date: 2024-12-31.
- **Modeling path:** BOAMP only. DECP is documented but not linked (no shared key between the two sources).

2.2 Corpus numbers

- 3,181 BOAMP notices downloaded (2015–2024).
- 1,933 are `APPEL_OFFRE` (source contracts); 1,081 are `ATTRIBUTION`.
- 1,210 are *eligible*: their expected renewal window (± 6 months around the estimated end date) closes before 2024-12-31.
- 723 are structurally censored upfront (too recent to observe a renewal).

DECP (3,039 digital contracts, same scope) is far richer (100% SIRET, 99.4% durations) but cannot be merged to BOAMP at contract level; contract-level DECP merging remains a future path.

2.3 Key data quality issues (modeling consequences)

Table 1: Key BOAMP data-quality findings and their modeling consequence. Full field-level detail is in the data quality report.

Finding	Value	Consequence
Buyer SIRET coverage (raw)	~9% of corpus	Grouping uses normalised names; SIREN enrichment later consolidates many (see §3.2).
Amount fill rate (APPEL_OFFRE)	~21%	Amount excluded from linking signals.
Declared duration granularity	69% at 12/24/36/48 mo	Duration used as a covariate, not as survival target.
Imputed durations	25% of eligible cohort	Flagged with <code>dur_was_imputed</code> ; imputation is 48 months.
No explicit renewal label	structural	<code>event</code> is a proxy reconstructed by the linking algorithm.

2.4 Taxonomy

Each contract is tagged with one of eleven technology categories (Software & Applications, IT Services & Consulting, Telecom & Networks, Cybersecurity, Cloud & Infrastructure, Data & AI, etc.) via a **deterministic rule-based mapping** from CPV codes and description keywords. This is *not* a trained classifier — the planned NLP classifier was not implemented. Output: `data/processed/taxonomy.csv`.

3 Main Pipeline

The pipeline runs end to end from raw BOAMP notices to operational risk indicators. Table 2 summarises each component and where its outputs live; the subsections below describe what each stage does, and Section 4 reports the numbers.

Table 2: Component status and key outputs. Status: **Done**, **Partial**, **Not started**.

Component	Status	Key result	Where
Data cleaning & EDA	Done	3,181 notices → 1,933 source contracts	data/processed/boamp_full_clean.csv
Buyer SIREN enrichment	Done	724/1,210 eligible rows on a SIREN key	buyer_siren_enrichment/
Renewal linking (event reconstruction)	Done	665/1,210 eligible linked (55.0%)	boamp_renewal_linking_quality/
Survival dataset construction	Done	1,210 rows × 33 cols	data/processed/boamp_phase2_survival.csv
Kaplan–Meier & log-rank	Done	Median 50.1 mo; duration significant	notebooks/02_survival_modeling_boamp.ipynb
Cox PH model	Done	C-index 0.6317	same notebook + reports/tables/survival/
Parametric AFT models	Done	Log-normal best (AIC 7,068.4)	same notebook
Sensitivity analysis	Done	Directions stable (A/B); strict variant underpowered	reports/tables/survival/sensitivity_comparison.csv
Robustness checks	Done	Placebo, bias check, threshold sweep	validation_robustness/
Operational risk indicators	Done	12/24-mo risk at contract/buyer/segment	reports/tables/survival/top20_renewal_risk.csv
Manual event validation	Done	150 cases audited; precision ≈0.09–0.15 (HIGH tier 0.50)	event_validation/manual_validation_summary.md
Trained NLP classifier	Partial	Scaffolding only: annotation sample (350 rows) + training scripts exist; gold labels and training not yet done	scripts/nlp_*.py, notebooks/03_nlp_classification.ipynb
Change-point detection (Phase 4)	Not done	Scoped but not implemented	—

3.1 BOAMP cleaning

Raw notices are parsed from the BOAMP API, deduplicated, and standardised: buyer key construction, contract start-date refinement (using `ATtribution` back-references where available), duration cleaning and imputation (missing eligible durations set to 48 months, flagged), CPV normalisation, and text normalisation. Output: `data/processed/boamp_full_clean.csv`.

3.2 Buyer SIREN enrichment

Because raw SIRET coverage is weak (~9% of the corpus), buyers were initially grouped by normalised name, which fragments a single public entity across spelling variants. A **SIREN enrichment step** (completed **2026-06-24**) queries the official *API Recherche d'Entreprises* for each buyer name and, for **HIGH-confidence matches only**, upgrades the buyer key from `NAME:` to `SIREN:{9-digit}`, merging name aliases of the same legal entity. Of the 1,210 eligible contracts, **724 carry a SIREN key (plus 2 SIRET) and 484 remain name-keyed**; 275 of 525 unique buyers (52.4%) matched with HIGH confidence. Medium/low matches are kept for review but never applied automatically. Enrichment is a buyer-resolution improvement, *not* ground truth: rare false merges at HIGH confidence remain possible. Outputs: `buyer_siren_enrichment/` and `data/processed/boamp_phase2_survival_siren_enriched.csv`.

3.3 Renewal-event reconstruction

Each eligible source contract is matched against later BOAMP notices from the same buyer using a four-signal composite score: semantic text embeddings (Sentence-Transformer, `paraphrase-multilingual-MiniLM-L12-v2`), CPV hierarchy match, temporal proximity, and buyer-identity agreement (weights 0.40 / 0.25 / 0.20 / 0.15). The best candidate passing the hard text-similarity filter is selected as the renewal successor and sets `event = 1`; contracts with no qualifying successor before study end are right-censored (`event = 0`). See technical report §5 for the full formula.

3.4 Survival modeling

The survival dataset (`data/processed/boamp_phase2_survival.csv`) has one row per eligible contract with observed/censored duration and covariates (`declared_duration_months`, `dur_was_imputed`, `start_year`, category, procedure type). Three model families were fitted in `notebooks/02_survival_modeling_boamp.ipynb`: Kaplan–Meier (with log-rank tests), a Cox proportional-hazards model, and parametric AFT models (Weibull, log-logistic, log-normal). The fitted model was stress-tested with a sensitivity analysis over three event definitions and a battery of robustness checks.

3.5 Operational risk indicators

The best parametric model (log-normal AFT) is applied to all scored contracts to produce conditional renewal probabilities at 12 and 24 months ahead, aggregated to contract, buyer, and segment level. These are prioritisation indicators, not legally verified renewal probabilities. Output: `reports/tables/survival/top20_renewal_risk.csv`.

4 Main Results

4.1 Renewal linking

Table 3: Renewal-linking outcome.

Quantity	Value	
Source contracts (<code>APPEL_OFFRE</code>)	1,933	
Eligible (renewal window observable)	1,210	
Linked renewals (<code>event</code> = 1)	665	(55.0%)
Right-censored (<code>event</code> = 0)	545	(45.0%)
Lexical baseline linking rate ($W = 6$)	7.6%	

Linking rate by category (Table 4): Software & Applications and IT Services & Consulting renew most (62.8%, 60.5%); Data & AI and IT Hardware & Equipment least (30.0%, 34.6%). Small categories ($n < 10$) are underpowered.

Table 4: Event rate by technology category (1,210 eligible contracts; events sum to 665).

Category	n	events	rate %
IT Services & Consulting	344	208	60.5
Software & Applications	223	140	62.8
Telecom & Networks	223	116	52.0
Cybersecurity	136	73	53.7
Unknown	115	49	42.6
Digital Workplace & Collaboration	51	32	62.7
Data & AI	40	12	30.0
IT Maintenance & Support	22	11	50.0
IT Hardware & Equipment	26	9	34.6
Cloud & Infrastructure	22	11	50.0
GIS & Mapping	8	4	50.0

Link quality. Of the 665 accepted links: 164 are single-candidate (no competing match); of the remaining 501 multi-candidate links, 219 (44%) have a score margin below 0.05, flagging an ambiguous winner. 75 links (11.3%) satisfy the `high_confidence_strict` criterion ($S \geq 0.70$ and margin ≥ 0.05 , multi-candidate only); these form the Model C core (Table 7). **Main unlinked failure mode:** 491 of 545 unlinked contracts (90.1%) have no temporal partner in the search window — a true absence of a candidate notice, not a scoring failure.

4.2 Survival analysis

Kaplan–Meier. Global median survival is **50.1 months**, consistent with the prevalence of 48-month framework contracts. Log-rank tests (Table 5): renewal timing is driven by declared duration and imputation status ($p < 0.001$); category is marginally significant ($p = 0.025$).

Table 5: Log-rank tests for differences in renewal timing.

Stratifying variable	statistic	p	df
Category (multivariate)	12.89	0.025	5
Duration was imputed	19.96	< 0.001	1
Declared-duration group	41.05	< 0.001	2
Procedure type (collapsed)	3.48	0.176	2

Cox proportional-hazards model. In-sample concordance: **0.6317**. Two predictors are significant (Table 6): longer declared duration lowers hazard (HR 0.991 per month), and later start year raises it (HR 1.071 per year). IT Services & Consulting shows a positive category effect (HR 1.230, $p = 0.037$). The positive `start_year` hazard is partly structural: 2021–2024 contracts are right-censored because a typical 48-month contract would renew after the 2024-12-31 study close.

Table 6: Cox model: significant covariates (multivariate).

Covariate	HR	95% CI	p
<code>declared_duration_months</code>	0.991	[0.986, 0.995]	< 0.001
<code>start_year</code>	1.071	[1.031, 1.113]	< 0.001
IT Services & Consulting (vs. Cybersecurity)	1.230	[1.013, 1.493]	0.037
<code>dur_was_imputed</code>	0.871	[0.722, 1.049]	0.146
Other category / procedure terms	not significant ($p > 0.1$)		

Parametric AFT models. Log-normal fits best (AIC 7,068.4, concordance 0.655), outperforming log-logistic (7,154.7) and Weibull (7,240.3). Full results: `reports/tables/survival/parametric_aic_comparison.csv`.

Sensitivity to event definition. Three event definitions were tested (Table 7). The Cox concordance drops from 0.632 (Model A) to 0.616 (Model B, ≥ 0.50) and 0.455 (Model C, strict H.C.), suggesting that low-confidence links add predictive signal not present in the strict subset — and that the 75-event strict definition is too sparse for stable estimation.

Table 7: Sensitivity to event definition. “—” = KM median never reached. KM median in months.

Variant	events	rate %	KM median	Cox C
A — baseline (all links)	665	55.0	50.1	0.632
B — conservative (≥ 0.50)	400	33.1	—	0.616
C — strict H.C. ($S \geq 0.70$, $\Delta \geq 0.05$, multi-cand.)	75	6.2	—	0.455

Proportional-hazards assumption. The Schoenfeld test flags a violation for `declared_duration_months` ($p < 10^{-5}$); all other covariates pass. Stratification or a time interaction for duration is future work. The log-normal AFT avoids the PH assumption and gives the same conclusions.

Temporal hold-out. A temporal validation (train on contracts starting ≤ 2021 : 1,088 rows / 604 events; test on ≥ 2022 : 116 rows / 60 events) gives a test concordance of 0.543 against 0.612 in-sample — a directional check only, since the small test cohort is structurally dominated by early renewals, but it confirms the in-sample C-index is an upper bound.

4.3 Operational risk indicators

The table below is computed under the *pre-calibration* baseline event definition (665 events). Under the calibrated balanced rule (269 events; see the calibration update on p. 1), the re-scored indicators are substantially lower: mean 12/24-month probabilities 0.023/0.068, max 12-month probability 0.073, and 27.5 expected recurrences within 12 months (83% lower than the 89.7 below). Files: `reports/tables/survival/*_calibrated_balanced.csv`.

Table 8: Operational risk summary (1,204 scored contracts).

Metric	Value
Mean 12-month renewal probability	0.074
Mean 24-month renewal probability	0.216
Max 12-month probability (any contract)	0.288
Contracts in <i>medium</i> band (0.25–0.50)	6
Contracts in <i>low</i> band (< 0.25)	1,198
Expected renewals within 12 months	89.7
Expected renewals within 24 months	259.9
Top buyer by expected 12-month volume	SIREN:234400034 (9.3)
Top segment by expected 12-month volume	IT Services & Consulting (30.4)

5 Reliability and Limitations

5.1 Reliability evidence

The event variable was stress-tested with checks in `validation_robustness/`:

- **Threshold sensitivity (7 scenarios: baseline, composite floors 0.50–0.90, strict H.C.):** Cox concordance declines gradually from 0.632 (baseline) to 0.599 (score ≥ 0.60), then becomes unstable below ≈ 100 events (0.473 at ≥ 0.70); hazard-ratio directions stay stable at every threshold with enough events. Full table: `validation_robustness/outputs/threshold_sensitivity_summary.csv`.
- **Placebo check:** winning-link mean score 0.554 vs. runner-up 0.428 vs. synthetic no-buyer 0.401 — the scoring discriminates (14.9% of winners reach ≥ 0.70 vs. 0.7% of runner-ups).
- **Event/censoring bias:** SMD = 0.17 (duration), -0.15 (start year) between event and censored groups — covariates reasonably balanced (SMD < 0.20); the dominant bias is structural right-censoring of 2022–2024 contracts.
- **Manual validation (executed 2026-07-02):** 150 stratified cases audited against the full official BOAMP records (`event_validation/outputs/manual_validation_audit_labeled.csv`). Result: TP 14 / FP 82 / uncertain 4 on 100 linked pairs (precision ≈ 0.15 raw, ≈ 0.09 weighted to the 665-event population; 0.50 in the HIGH/strict tier; 1.00 at text similarity ≥ 0.80); 6 of 47 decided unlinked sources hid a plausible missed renewal (12.8%). This is a sample-based audit estimate, not population ground truth; it invalidates the LOW tier and weakens the MEDIUM tier as renewal evidence.

5.2 Key limitations

- **Proxy event.** `event = 1` means a plausible renewal link was found, not a legally certified renewal. By composite score, 39.8% of the 665 links are below 0.50 (low), 45.3% between 0.50 and 0.70 (medium), and 14.9% at or above 0.70 (high).
- **Residual buyer fragmentation.** SIREN enrichment consolidated 724 of the 1,210 eligible rows onto a SIREN key (plus 2 SIRET), but 484 remain name-keyed, so spelling variants not resolved by the API can still create false negatives. Enrichment is not ground truth.
- **Short-text false positives.** Generic descriptions (“prestations informatiques”) can match spuriously.
- **Structural right-censoring.** Contracts from 2021–2024 are largely censored; the positive `start_year` hazard partly reflects study truncation, not a genuine cohort trend.

- **Modest discrimination.** Concordance ≈ 0.65 — useful for prioritisation, not sufficient for high-stakes individual decisions.

6 Completed Work and Remaining Work

Completed. BOAMP acquisition and cleaning; buyer SIREN enrichment (2026-06-24); pre-calibration renewal-event reconstruction (665 events, 55.0%); synthetic benchmark calibration with the real Sentence-Transformer encoder; calibrated balanced event definition (269 events, 22.2%); calibrated survival rerun (KM, Cox, AFT); operational 12/24-month risk indicators (pre-calibration and re-scored calibrated versions); rule-based technology taxonomy.

Remaining.

- **Use the calibrated event definition.** The 150-case audit (executed 2026-07-02) showed that the broad baseline was too permissive. The priority is now to use the balanced calibrated dataset as the main survival input and keep broad/strict variants as sensitivity checks.
- **Trained NLP classifier.** The technology taxonomy is rule-based (CPV + keywords). The classifier pipeline is scaffolded (`scripts/nlp_annotation_sample.py` through `scripts/nlp_propagate_labels.py` and notebook 03), and the 350-row annotation sample is drawn, but the gold labels have not been filled in and no model has been trained yet.
- **Change-point / trend detection (Phase 4).** Scoped but not started.
- **DECP semantic linking.** Applied to BOAMP only; DECP linking remains at the lexical prototype stage.
- **Live prediction scoring.** The current model evaluates contract age as of the study end date; a production run would re-evaluate each contract at a live prediction date.

7 Conclusion and Next Steps

The internship delivered a reproducible end-to-end pipeline: BOAMP notice acquisition → cleaning → buyer SIREN enrichment → renewal-event reconstruction → survival modeling → operational risk indicators. The main methodological result is a calibrated proxy-event definition: the broad Sentence-Transformer rule gives 490 events (40.5%), the balanced rule gives 269 events (22.2%), and the strict rule gives 79 events (6.5%). Under the balanced rule, the Kaplan–Meier median is not reached and LogNormalAFT is the best AFT model by AIC (3,544.8). These outputs translate BOAMP proxy recurrence signals into 12/24-month renewal-risk rankings.

Immediate next steps (prioritised):

1. **Run the next survival interpretation on the balanced dataset.** Use broad and strict only as sensitivity checks; do not report real precision/recall without manual labels.
2. **Re-score contracts at a live prediction date** (current scores use end-of-study ages).
3. **Review medium/low SIREN matches** to extend buyer consolidation beyond the 724 HIGH-confidence rows already applied.
4. **Replace the rule-based taxonomy** with a trained text classifier.
5. **Implement change-point detection** (Phase 4).